

H&E + arm-level CNV fusion for PAM50 subtype: internal and external evaluation summary

dp-code project (compiled from on-disk results)

2026-09-01

Abstract

Does conditioning H&E whole-slide images on arm-level copy number beat averaging two independent predictions for four-class PAM50 subtyping? No. Trained on TCGA-BRCA and validated on CPTAC-BRCA (114 cases, 378 slides, never trained on), the untrained equal-weight probability mean of a CLAM-MB / UNI2-h WSI model and a 39-arm copy-number logistic regression reaches macro AUROC 0.909 [0.861, 0.949] externally, against 0.847 for WSI alone and 0.888 for CNV alone; a post hoc prior-balanced mean reaches 0.912. Internally, on 599 pooled out-of-fold TCGA cases with the CNV arm refit per CLAM fold, the mean reaches 0.926 against WSI 0.887 and CNV 0.872, and all five jointly trained fusion operators score 0.882 to 0.899, every one significantly below that untrained mean. CNV alone is reported with every fusion number: externally the equal-weight mean is not distinguishable from it.

1 Setup

Task and unit. Four-class PAM50 subtype (Luminal A, Luminal B, Basal-like, HER2-enriched); Normal-like is dropped and CPTAC contains no Normal-like case, so the cohorts share a label space. The prediction unit is the case: slide-level probabilities are mean-pooled to the case before any metric is computed. Macro AUROC is one-vs-rest.

Table 1: Case sets. Each TCGA row is a subset of the one above it. Source: `data/cohort_counts.csv`.

Case set	LumA	LumB	Basal	Her2	Normal	Total
TCGA labelled cases (incl. Normal-like)	499	197	171	78	36	981
TCGA non-Normal with arm CNV (the CNV fitting set)	499	197	171	78	0	945
TCGA CLAM tabular table (CNV + WSI + fold)	475	195	165	75	0	910
TCGA with CLAM out-of-fold predictions (all internal contrasts)	318	124	106	51	0	599
CPTAC-BRCA external cases (378 slides)	56	17	27	14	0	114

Arms. The WSI arm is UNI2-h 1536-dimensional patch features aggregated by CLAM-MB (`model_size` big, dropout 0.5, inverse-frequency weighted sampler, so no claim of “natural class frequencies” applies), 10 folds at seed 1 on the tracked split set `splits/tcga_brca_subtyping_100`. The CNV arm is 39 chromosome arms, each feature the median gene-level $\log_2$ ratio over that arm, which is the scale deployed low-pass assays report; its classifier is a frozen constant defined once in `tools/pam50_arms.py`, `StandardScaler` $\rightarrow$ `LogisticRegression(max_iter=4000, C=0.1, class_weight='balanced')`, refit in process and never stored as a checkpoint. Copy number rather than RNA is the second modality because the PAM50 labels are computed from the expression matrix itself, so an RNA branch leaks the target by construction; the project’s WSI+RNA number is inflated for that reason and is not quoted here.

Fusion. The untrained *equal-weight probability mean* averages the two four-class probability vectors element-wise; it has no parameters, was fixed on TCGA before CPTAC was scored, and is the baseline every trained operator must clear. Five trained operators (`concat`, `gated`, `cross_attention`, `film_attention`, `coattn`) pair the CLAM-MB branch with a two-layer tabular encoder over the 39 arms (`tabular_hidden_dim` 64, `fusion_hidden_dim` 32), all warm-started from the corresponding fold’s WSI-only checkpoint and not frozen; `film_attention` conditions the attention network’s input through a rank-16 FiLM bottleneck with 0.2 modality dropout, and `coattn` runs co-attention between patch tokens and 22 chromosome tokens. A sixth operator, `residual`, refuses at composition time because no supported trainer produces its tabular checkpoint, so the ladder is five operators, not six.

Protocols and frozen constants. Internal metrics are pooled out-of-fold: held-out predictions concatenated over the ten folds and scored once, a different quantity from the mean of ten per-fold AUCs in CLAM’s

`summary.csv` (0.893 there against 0.887 pooled; never mix them). Internal contrasts use a paired bootstrap of $N = 2000$ at seed 13 with the CNV arm refit per CLAM fold; external contrasts use $N = 4000$ at seed 7; a third bootstrap, $N = 2000$ at seed 11, belongs to the stacking analysis and is not quoted here. The three seeds differ deliberately and unifying them would move published intervals. Resamples that lose a class are rejected so macro AUROC stays defined. Two class orders exist, CLAM’s LumA, LumB, Basal, Her2 and pam50_arms.CLASSES sorted Basal, Her2, LumA, LumB; they are bridged by `clam_column_order()` before anything is scored across them, and every table here is in the sorted order.

2 Results

Table 2: PAM50 four-class subtype classification, H&E WSI and arm-level CNV, on TCGA-BRCA (internal, pooled out-of-fold) and CPTAC-BRCA (external, never trained on). The untrained equal-weight probability mean is the baseline in both cohorts; every trained fusion operator falls significantly below it internally, and no trained operator has an external evaluation path. CNV alone is reported alongside fusion throughout.

Arm	Macro AUROC [95% CI]	Balanced acc. [95% CI]	HER2 recall (k/n)	Δ AUROC vs. mean [95% CI]
Internal: TCGA-BRCA, 10-fold pooled out-of-fold, $n = 599$				
WSI only (CLAM-MB, UNI2-h)	0.887 [0.867, 0.909]	0.677 [0.631, 0.725]	26/51	−0.039 [−0.054, −0.023] *
CNV only (39 arms)	0.872 [0.848, 0.895]	0.678 [0.632, 0.724]	25/51	−0.054 [−0.073, −0.037] *
Equal-weight probability mean (baseline, untrained)	0.926 [0.910, 0.941]	0.751 [0.711, 0.794]	30/51	—
<i>Trained fusion operators (joint WSI+CNV training, warm-started from the WSI baseline)</i>				
<code>concat</code>	0.883 [0.859, 0.906]	0.674 [0.629, 0.721]	n/a ^e	−0.043 [−0.063, −0.026] *
<code>gated</code>	0.895 [0.875, 0.915]	0.683 [0.637, 0.730]	n/a ^e	−0.031 [−0.048, −0.016] *
<code>cross_attention</code>	0.892 [0.869, 0.914]	0.685 [0.639, 0.730]	n/a ^e	−0.034 [−0.054, −0.016] *
<code>film_attention</code>	0.882 [0.861, 0.903]	0.665 [0.618, 0.714]	n/a ^e	−0.044 [−0.062, −0.027] *
<code>coattn</code>	0.899 [0.879, 0.919]	0.684 [0.639, 0.730]	n/a ^e	−0.027 [−0.043, −0.011] *
External: CPTAC-BRCA, never trained on, $n = 114$ (378 slides)				
WSI only (raw)	0.847 [0.793, 0.897]	0.513 [0.455, 0.579]	0/14	−0.063 [−0.104, −0.024] *
CNV only (39 arms)	0.888 [0.833, 0.935]	0.716 [0.622, 0.805]	12/14	−0.021 [−0.047, +0.003] ^{ns}
Equal-weight probability mean (baseline, untrained)	0.909 [0.861, 0.949]	0.646 [0.548, 0.746]	6/14	—
<i>Post hoc on CPTAC (†)</i>				
WSI only, prior-balanced [†]	0.865 [0.813, 0.911]	0.554 [0.484, 0.623]	0/14	−0.045 [−0.083, −0.008] *
Prior-balanced mean (WSI + CNV) [†]	0.912 [0.864, 0.952]	0.740 [0.638, 0.832]	10/14	+0.003 [−0.005, +0.011] ^{ns}

* / ^{ns}: the 95% bootstrap interval on the paired difference excludes / includes zero. Δ is the arm minus the equal-weight probability mean of its own cohort, so a negative value is worse than the untrained baseline.

† Post hoc: computed on CPTAC after inspection; no CPTAC tuning of any model. Nothing in this work was refit, calibrated or thresholded on CPTAC.

^a Protocols. Internal: 10 CLAM folds pooled out-of-fold on the 599 TCGA cases that reach a test fold, the CNV arm refit per CLAM fold, paired bootstrap $N = 2000$ at seed 13. External: TCGA-trained models applied to CPTAC untouched, paired bootstrap $N = 4000$ at seed 7. The two bootstrap seeds differ deliberately and are not unified.

^b The equal-weight probability mean is *untrained*: it averages the two four-class probability vectors element-wise and has no parameters. It is the baseline, not the WSI-only model.

^c No trained fusion operator has an external evaluation path: `dp-evaluate` refuses `film_attention` and `coattn` checkpoints by name and defaults to the TCGA test split, so the five operators are internal-only and every external fusion row above is the untrained probability mean of two independently trained arms.

^d CLAM’s 10 splits are drawn independently rather than partitioned: 599 of 910 cases reach a test fold and 242 of them are tested by two to five models, so the pooled WSI-only figure is a small ensemble, flattered by roughly +0.01 macro AUROC. All six trained internal runs share the split set, so the internal ladder comparison stays self-consistent.

^e Per-class recall for the five trained operators is not produced by any script in the repository and exists nowhere on disk, hence n/a.

^f The CNV arm *as a model in its own right* scores macro AUROC 0.866 ± 0.003 (5-fold CV averaged over 10 reseeds, 945 TCGA non-Normal cases; the $\pm$ is the spread over reseeds, not a bootstrap CI), and aneuploidy burden alone (one feature, mean absolute $\log_2$, 5-fold at seed 0, the published protocol) reaches 0.685 on the same set. The internal 0.872 above is the per-CLAM-fold refit; the same 599 cases under `StratifiedKFold(10, seed 0)` give 0.862 with an equal-weight mean of 0.922, so name the protocol whenever an internal CNV number is quoted.

Internal (TCGA-BRCA, 599 pooled out-of-fold cases). The untrained equal-weight mean reaches macro AUROC 0.926 [0.910, 0.941] and balanced accuracy 0.751, above WSI alone (0.887) and CNV alone (0.872, refit per CLAM fold). All five trained operators land between 0.882 (`film_attention`) and 0.899 (`coattn`), and each is significantly below the mean on a paired bootstrap, by −0.027 to −0.044 macro AUROC; no trained operator reaches even the better unimodal arm plus the gap that averaging buys. HER2 recall moves with AUROC: 26/51 for WSI, 25/51 for CNV, 30/51 for the mean. The mechanism is error correlation,

quantified in Section 3.

External (CPTAC-BRCA, 114 cases, never trained on). The ordering is preserved and the HER2 story changes completely. WSI alone reaches 0.847 with HER2 recall 0/14, not one of the fourteen HER2-enriched cases recovered. CNV alone reaches 0.888 with 12/14, and the equal-weight mean 0.909 with 6/14, the mean inheriting the WSI arm’s confident errors on that class. The post hoc prior-balanced mean reaches 0.912 with 10/14 and the best balanced accuracy (0.740), but it is post hoc and its AUROC gain over the equal-weight mean is not significant. CNV alone remains the arm to beat: the equal-weight mean does not separate from it, and only the post hoc prior-balanced mean does, with a lower bound that prints as +0.000. No trained operator appears in the external block because none can be evaluated externally.

Table 3: External per-class AUROC and recall on CPTAC-BRCA, $n = 114$, support Basal 27 / HER2 14 / LumA 56 / LumB 17. Protocol as in Table 2 ($N = 4000$, seed 7); † marks post hoc arms (computed on CPTAC after inspection, no CPTAC tuning of any model). Source: `data/external_per_class.csv`.

Arm	AUROC (one-vs-rest)				Recall (k/n)			
	Basal	HER2	LumA	LumB	Basal	HER2	LumA	LumB
WSI only (raw)	0.972	0.860	0.861	0.693	24/27	0/14	52/56	4/17
CNV only (39 arms)	0.972	0.871	0.883	0.827	22/27	12/14	37/56	9/17
Equal-weight mean (baseline)	0.992	0.881	0.916	0.848	23/27	6/14	50/56	7/17
WSI only, prior-balanced†	0.967	0.869	0.863	0.760	26/27	0/14	34/56	11/17
Prior-balanced mean†	0.990	0.893	0.918	0.848	24/27	10/14	43/56	10/17

Basal-like is near-saturated for every arm (0.967 to 0.992) and is the class most exposed to the aneuploidy-burden control: one genome-instability feature already reaches 0.685 macro AUROC on the 945-case fitting set (5-fold at seed 0, the published protocol), so Basal ≈ 0.97 must be read beside that number rather than as evidence that the arm pattern does all the work. LumB is the weakest class everywhere and, at $n = 17$, the least certain.

3 Key contrasts

Deltas are paired bootstraps on the same case set; *sig* means the 95% interval excludes zero. Internal: $N = 2000$, seed 13, CNV refit per CLAM fold. External: $N = 4000$, seed 7. Source: `data/contrasts.csv`.

- **Internal, mean vs. WSI alone:** AUROC +0.039 [+0.023, +0.054] sig; balanced accuracy +0.074 [+0.027, +0.121] sig.
- **Internal, mean vs. CNV alone:** AUROC +0.054 [+0.037, +0.073] sig; balanced accuracy +0.073 [+0.033, +0.116] sig.
- **Internal, each trained operator vs. the mean:** negative and sig throughout, from -0.027 [-0.043 , -0.011] (coattn, best) to -0.044 [-0.062 , -0.027] (film_attention, worst); balanced accuracy sig-negative for all five.
- **External, equal-weight mean vs. WSI alone:** +0.063 [+0.024, +0.104] sig.
- **External, equal-weight mean vs. CNV alone:** +0.021 [-0.003 , +0.047], *ns*. Averaging does not measurably improve on copy number alone externally.
- **External, post hoc prior-balanced mean vs. CNV alone:** +0.024 [+0.000, +0.050] sig, the lower bound exactly +0.000 at three decimals (full precision +0.00047). This is the whole edge of fusion over CNV alone, and it is post hoc.
- **External, post hoc prior-balanced mean vs. equal-weight mean:** AUROC +0.003 [-0.005 , +0.011] *ns*; balanced accuracy +0.093 [+0.014, +0.177] sig.
- **External, CNV alone vs. WSI alone:** AUROC +0.042 [-0.019 , +0.103] *ns*; balanced accuracy +0.202 [+0.089, +0.315] sig.
- **Error correlation ϕ between per-case errors.** Among the five trained operators, mean over the ten operator pairs, internal pooled out-of-fold: $\phi = 0.656$ (0.582 to 0.706). Between the two unimodal arms, internal, CNV refit per CLAM fold: $\phi = 0.193$; the same pair under StratifiedKFold(10, seed 0), which is the published value: $\phi = 0.269$. Externally, raw WSI vs. CNV: $\phi = -0.006$, either arm right on

91.2% of cases while each alone is right on 70.2%. Joint training collapses the diversity that averaging exploits.

For context, and as the only comparable published work pairing copy number as the sole molecular modality with H&E for PAM50, Amer 2025 reports CNV+Image 0.8835 against CNV alone 0.8284, and its own untrained “Simple Ensemble” at 0.9074 beats its trained operators. The pattern here is the same one, extended to an external cohort and reported alongside the CNV-alone arm.

4 Provenance

Every number above comes from the artifacts in Table 4 and from the files copied into `data/` beside this document. The 2026-09-01 recompute reproduced the 2026-08-06 runs with zero discrepancies above 0.001 over 108 paired checks. The `dp-*` console scripts are not on PATH in this container, so each 2026-09-01 command was invoked as `/opt/venv/bin/python -m dpcode.cli.analysis <action> [overrides]`, the same `main()` the console script calls, and each run’s `run_metadata.json` records the canonical `dp-analysis` form. `dp-analysis` exposes `--show-config`, not `--dry-run`; the dry-run step used `--show-config`.

Table 4: Artifacts behind this summary, the command that produced each, and its date on disk. Paths are absolute; everything under `.scratch/` and `.datasets/` is gitignored and not recoverable from git.

Artifact	Producing command	Date
<code>/workspace/dp-code/.scratch/results/pam50_final_s1/</code> WSI baseline; <code>split_*_results.pkl</code> is the internal WSI arm, <code>s_*_checkpoint.pt</code> the ladder’s warm start	<code>dp-train experiment=pam50_wsi_final</code>	2026-03-27
<code>/workspace/dp-code/.scratch/results/pam50_wsi_cnv_concat_s1/</code> and the sibling <code>_gated_s1</code> , <code>_cross_attention_s1</code> , <code>_film_attention_s1</code> , <code>_coattn_s1</code> dirs	<code>dp-train -m experiment=pam50_wsi_cnv fusion=concat,gated,cross_attention,film_attention,coattn</code>	2026-08-05
<code>/workspace/dp-code/.datasets/cnv/tcga_brca_cna_arm.csv</code> , .../ <code>cptac_brca_cna_arm.csv</code> , .../ <code>reference/gene_arm_hg38.csv</code> (the only pin on the 39-arm feature space)	<code>dp-data cnv</code>	2026-08-03
<code>/workspace/dp-code/.scratch/cnv-tabular/TCGA_BRCA_CNV_arm_4class_clam.csv</code> (910 rows), .../ <code>CPTAC_BRCA_CNV_arm_4class_clam.csv</code> (114), .../ <code>chromosome_groups.csv</code>	<code>dp-analysis make_cnv_tabular</code>	2026-08-05
<code>/workspace/dp-code/.scratch/cptac_validation/results/predictions/ensemble_predictions.csv</code> (378 slides, 114 cases; the external WSI arm)	<code>dp-cptac phase=3 (tools/cptac/infer_cptac_pam50.py)</code>	2026-07-29
<code>/workspace/dp-code/.scratch/analysis/cnv_wsi_fusion/2026-09-01_18-58-28/</code> and .../ <code>cnv_wsi_fusion/2026-09-01_18-59-16/</code>	<code>dp-analysis cnv_wsi_fusion</code> (external), then the same with <code>analysis.internal=true</code>	2026-09-01
<code>/workspace/dp-code/.scratch/analysis/stack_wsi_cnv/2026-09-01_19-00-36/</code>	<code>dp-analysis stack_wsi_cnv</code>	2026-09-01
<code>/workspace/dp-code/.scratch/analysis/compare_fusion_ladder/2026-09-01_19-01-21/</code>	<code>dp-analysis compare_fusion_ladder</code>	2026-09-01
<code>/workspace/dp-code/.scratch/analysis/cnv_controls/2026-09-01_19-02-20/</code> and .../ <code>cnv_controls/2026-09-01_19-04-48/</code>	<code>dp-analysis cnv_controls</code> , then the same with <code>analysis.burden_definition=frac_altered</code>	2026-09-01
<code>/workspace/dp-code/.scratch/analysis/cnv_wsi_fusion/2026-08-06_10-38-00/</code> , .../ <code>stack_wsi_cnv/2026-08-06_10-39-11/</code> , .../ <code>cnv_controls/2026-08-06_10-39-57/</code> , .../ <code>compare_fusion_ladder/2026-08-06_10-40-00/</code> (the runs the 2026-09-01 set reproduces)	<code>dp-analysis cnv_wsi_fusion analysis.internal=true;</code> <code>dp-analysis stack_wsi_cnv; dp-analysis cnv_controls;</code> <code>dp-analysis compare_fusion_ladder</code>	2026-08-06
<code>/workspace/dp-code/tools/pam50_arms.py</code> (frozen CNV arm, every path constant, the bootstrap, the class-order bridge)	imported by every analysis script, the report build and the notebook	standing
<code>/workspace/dp-code/.scratch/wsi_cnv_report/</code> (CSVs, figures, the long report; the balanced-accuracy CIs and the full-precision per-class AUROC come from here)	.../ <code>build_csvs.py</code> , then <code>pdflatex</code> on <code>wsi_cnv_evaluation.tex</code> ; published copy <code>/workspace/dp-code/docs/wsi_cnv_evaluation.pdf</code>	CSVs 2026-08-06, PDF 2026-08-07
<code>/workspace/dp-code/tools/evaluate_pam50_fusion.ipynb</code> (supervisor report, recomputes every published number from disk; untracked)	run interactively; PDF at <code>/workspace/dp-code/evaluate_pam50_fusion.pdf</code>	2026-08-08
<code>/workspace/dp-code/docs/wsi_cnv_summary/data/</code> (<code>results_consolidated.csv</code> , <code>contrasts.csv</code> , <code>external_per_class.csv</code> , <code>eval_notes.md</code> , <code>cohort_counts.csv</code>)	copied unchanged from the 2026-09-01 evaluation and from .../ <code>wsi_cnv_report/</code>	2026-09-01

5 Caveats and limitations

- **The external HER2 collapse is domain shift, not calibration.** 0/14 survives a 12-fold prior boost and SLD-EM label shift (that arm, post hoc, gives 0.858 with recall still 0/14 and an implied HER2 prior of 0.000). The mechanism is UNI2-h feature-space geometry: cohort of origin is readable from a single patch feature at AUROC 1.0000, the cohort shift is 5.4 times the HER2 class signal, and a logistic probe on slide-mean features reproduces 0/378, which exonerates the MIL head. Aggregation, segmentation dilution, label mismatch and a 17q-specific explanation were refuted; CNV’s 12/14 holds without 17q.
- **Power.** External HER2 is $n = 14$ and LumB $n = 17$; per-class external numbers for those classes, and any recall expressed as $k/14$, are indicative only.
- **The ladder is a near-baseline, not a matched, comparison.** The WSI baseline uses sweep-selected optimiser values with `inst_loss svm` and instance clustering on; the five operators use rounded learning rate and weight decay, CLAM’s default bag weight 0.7 and `--no_inst_cluster`. Reproduced as run rather than tidied.
- **Shared warm start, unseparated confound.** All five operators warm-start from the same `pam50_final_s1` checkpoints, so joint-training collapse and shared-initialisation diversity collapse cannot be told apart from these runs; a `--no_warm_start` arm would settle it and has not been run.
- **CPTAC and TCGA features are geometrically comparable but not identically preprocessed** (CLAM tissue segmentation for TCGA, Trident for CPTAC).
- **The external cohort is stage IIA to IIIC with no Normal-like case.** External CNV 0.888 against internal 0.866 therefore means “does not degrade”, not “improves”: different case mixes under different protocols.
- **Two internal CNV protocols differ by about 0.010, so name the one in use.** 0.872 is the refit per CLAM fold (599 cases) used for every contrast here; 0.862 is `StratifiedKFold(10, seed 0)` on the same cases, whose equal-weight mean is 0.922 rather than 0.926; 0.866 ± 0.003 is the 5-fold by 10-reseed figure on the full 945-case fitting set and is the right one for the CNV branch as a model in its own right. The spread exceeds several contrasts called significant, which is why $\phi = 0.193$ (per-CLAM-fold) and $\phi = 0.269$ (StratifiedKFold) are always quoted with their protocol.
- **Independent splits, no external operator evaluation.** CLAM’s ten splits are drawn independently rather than partitioned, so pooled WSI-only is a small ensemble flattered by roughly +0.01 AUROC (footnote d of Table 2); and no trained operator can be evaluated externally (footnote c), so every external fusion row is the untrained probability mean.
- **Reproduction environment and determinism.** The console scripts are not installed on PATH here, so the 2026-09-01 runs used `python -m dpcode.cli.analysis <action>; pip install -e '[dev]'` restores them, and `dp-analysis` has `--show-config` rather than `--dry-run`. Nothing was trained, downloaded or overwritten for this summary. Seeds, folds and split sets are fixed, but run-to-run variance is unmeasured and bitwise reproducibility is not achievable in this stack.

The long-form report, with figures, per-fold tables, the stacking analysis and the calibration controls, is `/workspace/dp-code/docs/wsi_cnv_evaluation.pdf`. The project write-up of the external validation is `/workspace/dp-code/docs/cnv-wsi-fusion-external-validation.md` (author-owned).